

Chapter 8

Least-Squares Fitting

Our discussion of the statistical analysis of data has so far focused exclusively on the repeated measurement of one single quantity, not because the analysis of many measurements of one quantity is the most interesting problem in statistics, but because this simple problem must be well understood before more general ones can be discussed. Now we are ready to discuss our first, and very important, more general problem.

8.1 Data That Should Fit a Straight Line

One of the most common and interesting types of experiment involves the measurement of several values of two different physical variables to investigate the mathematical relationship between the two variables. For instance, an experimenter might drop a stone from various different heights $h_1, \dots, h_N$ and measure the corresponding times of fall $t_1, \dots, t_N$ to see if the heights and times are connected by the expected relation $h = \frac{1}{2}gt^2$.

Probably the most important experiments of this type are those for which the expected relation is *linear*. For instance, if we believe that a body is falling with constant acceleration g , then its velocity v should be a linear function of the time t ,

$$v = v_0 + gt.$$

More generally, we will consider any two physical variables x and y that we suspect are connected by a linear relation of the form

$$y = A + Bx, \tag{8.1}$$

where A and B are constants. Unfortunately, many different notations are used for a linear relation; beware of confusing the form (8.1) with the equally popular $y = ax + b$.

If the two variables y and x are linearly related as in (8.1), then a graph of y against x should be a straight line that has slope B and intersects the y axis at $y = A$. If we were to measure N different values $x_1, \dots, x_N$ and the corresponding values $y_1, \dots, y_N$ and if our measurements were subject to no uncertainties, then each of the points (x_i, y_i) would lie exactly on the line $y = A + Bx$, as in Figure 8.1(a). In

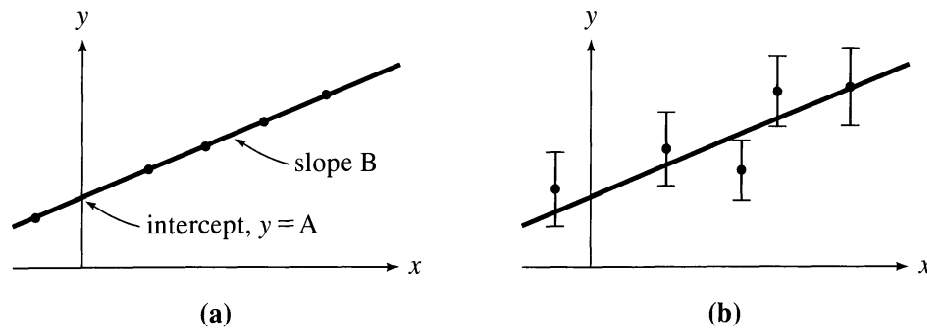


Figure 8.1. (a) If the two variables y and x are linearly related as in Equation (8.1), and if there were no experimental uncertainties, then the measured points (x_i, y_i) would all lie exactly on the line $y = A + Bx$. (b) In practice, there always are uncertainties, which can be shown by error bars, and the points (x_i, y_i) can be expected only to lie reasonably close to the line. Here, only y is shown as subject to appreciable uncertainties.

practice, there *are* uncertainties, and the most we can expect is that the distance of each point (x_i, y_i) from the line will be reasonable compared with the uncertainties, as in Figure 8.1(b).

When we make a series of measurements of the kind just described, we can ask two questions. First, if we take for granted that y and x *are* linearly related, then the interesting problem is to find the straight line $y = A + Bx$ that best fits the measurements, that is, to find the best estimates for the constants A and B based on the data $(x_1, y_1), \dots, (x_N, y_N)$. This problem can be approached graphically, as discussed briefly in Section 2.6. It can also be treated analytically, by means of the principle of maximum likelihood. This analytical method of finding the best straight line to fit a series of experimental points is called *linear regression*, or the *least-squares fit for a line*, and is the main subject of this chapter.

The second question that can be asked is whether the measured values $(x_1, y_1), \dots, (x_N, y_N)$ do really bear out our expectation that y is linear in x . To answer this question, we would first find the line that best fits the data, but we must then devise some measure of *how well* this line fits the data. If we already know the uncertainties in our measurements, we can draw a graph, like that in Figure 8.1(b), that shows the best-fit straight line and the experimental data with their error bars. We can then judge visually whether or not the best-fit line passes sufficiently close to all of the error bars. If we do not know the uncertainties reliably, we must judge how well the points fit a straight line by examining the distribution of the points themselves. We take up this question in Chapter 9.

8.2 Calculation of the Constants A and B

Let us now return to the question of finding the best straight line $y = A + Bx$ to fit a set of measured points $(x_1, y_1), \dots, (x_N, y_N)$. To simplify our discussion, we will suppose that, although our measurements of y suffer appreciable uncertainty, the uncertainty in our measurements of x is negligible. This assumption is often reasonable, because the uncertainties in one variable often are much larger than

those in the other, which we can therefore safely ignore. We will further assume that the uncertainties in y all have the same magnitude. (This assumption is also reasonable in many experiments, but if the uncertainties are different, then our analysis can be generalized to weight the measurements appropriately; see Problem 8.9.) More specifically, we assume that the measurement of each y_i is governed by the Gauss distribution, with the same width parameter σ_y for all measurements.

If we knew the constants A and B , then, for any given value x_i (which we are assuming has no uncertainty), we could compute the true value of the corresponding y_i ,

$$(\text{true value for } y_i) = A + Bx_i. \quad (8.2)$$

The measurement of y_i is governed by a normal distribution centered on this true value, with width parameter σ_y . Therefore, the probability of obtaining the observed value y_i is

$$\text{Prob}_{A,B}(y_i) \propto \frac{1}{\sigma_y} e^{-(y_i - A - Bx_i)^2 / 2\sigma_y^2}, \quad (8.3)$$

where the subscripts A and B indicate that this probability depends on the (unknown) values of A and B . The probability of obtaining our complete set of measurements $y_1, \dots, y_N$ is the product

$$\begin{aligned} \text{Prob}_{A,B}(y_1, \dots, y_N) &= \text{Prob}_{A,B}(y_1) \cdots \text{Prob}_{A,B}(y_N) \\ &\propto \frac{1}{\sigma_y^N} e^{-\chi^2/2}, \end{aligned} \quad (8.4)$$

where the exponent is given by

$$\chi^2 = \sum_{i=1}^N \frac{(y_i - A - Bx_i)^2}{\sigma_y^2}. \quad (8.5)$$

In the now-familiar way, we will assume that the best estimates for the unknown constants A and B , based on the given measurements, are those values of A and B for which the probability $\text{Prob}_{A,B}(y_1, \dots, y_N)$ is maximum, or for which the sum of squares χ^2 in (8.5) is a minimum. (This is why the method is known as least-squares fitting.) To find these values, we differentiate χ^2 with respect to A and B and set the derivatives equal to zero:

$$\frac{\partial \chi^2}{\partial A} = \frac{-2}{\sigma_y^2} \sum_{i=1}^N (y_i - A - Bx_i) = 0 \quad (8.6)$$

and

$$\frac{\partial \chi^2}{\partial B} = \frac{-2}{\sigma_y^2} \sum_{i=1}^N x_i (y_i - A - Bx_i) = 0. \quad (8.7)$$

These two equations can be rewritten as simultaneous equations for A and B :

$$AN + B \sum x_i = \sum y_i \quad (8.8)$$

and

$$A \sum x_i + B \sum x_i^2 = \sum x_i y_i. \quad (8.9)$$

Here, I have omitted the limits $i = 1$ to N from the summation signs Σ . In the following discussion, I also omit the subscripts i when there is no serious danger of confusion; thus, $\Sigma x_i y_i$ is abbreviated to Σxy and so on.

The two equations (8.8) and (8.9), sometimes called *normal equations*, are easily solved for the least-squares estimates for the constants A and B ,

$$A = \frac{\Sigma x^2 \Sigma y - \Sigma x \Sigma xy}{\Delta} \quad (8.10)$$

and

$$B = \frac{N \Sigma xy - \Sigma x \Sigma y}{\Delta}, \quad (8.11)$$

where I have introduced the convenient abbreviation for the denominator,

$$\Delta = N \Sigma x^2 - (\Sigma x)^2. \quad (8.12)$$

The results (8.10) and (8.11) give the best estimates for the constants A and B of the straight line $y = A + Bx$, based on the N measured points $(x_1, y_1), \dots, (x_N, y_N)$. The resulting line is called the *least-squares fit* to the data, or the *line of regression* of y on x .

Example: Length versus Mass for a Spring Balance

A student makes a scale to measure masses with a spring. She attaches its top end to a rigid support, hangs a pan from its bottom, and places a meter stick behind the arrangement to read the length of the spring. Before she can use the scale, she must calibrate it; that is, she must find the relationship between the mass in the pan and the length of the spring. To do this calibration, she gets five accurate 2-kg masses, which she adds to the pan one by one, recording the corresponding lengths l_i as shown in the first three columns of Table 8.1. Assuming the spring obeys Hooke's law, she anticipates that l should be a linear function of m ,

$$l = A + Bm. \quad (8.13)$$

(Here, the constant A is the unloaded length of the spring, and B is g/k , where k is the usual spring constant.) The calibration equation (8.13) will let her find any unknown mass m from the corresponding length l , once she knows the constants A and B . To find these constants, she uses the method of least squares. What are her answers for A and B ? Plot her calibration data and the line given by her best fit (8.13). If she puts an unknown mass m in the pan and observes the spring's length to be $l = 53.2$ cm, what is m ?

Table 8.1. Masses m_i (in kg) and lengths l_i (in cm) for a spring balance. The “x” and “y” in quotes indicate which variables play the roles of x and y in this example.

Trial number i	“x” Load, m_i	“y” Length, l_i	m_i^2	$m_i l_i$
1	2	42.0	4	84
2	4	48.4	16	194
3	6	51.3	36	308
4	8	56.3	64	450
5	10	58.6	100	586
$N = 5$	$\Sigma m_i = 30$	$\Sigma l_i = 256.6$	$\Sigma m_i^2 = 220$	$\Sigma m_i l_i = 1,622$

As often happens in such problems, the two variables are not called x and y , and one must be careful to identify which is which. Comparing (8.13) with the standard form, $y = A + Bx$, we see that the length l plays the role of the dependent variable y , while the mass m plays the role of the independent variable x . The constants A and B are given by (8.10) through (8.12), with the replacements

$$x_i \leftrightarrow m_i \quad \text{and} \quad y_i \leftrightarrow l_i.$$

(This correspondence is indicated by the headings “x” and “y” in Table 8.1.) To find A and B , we need to find the sums Σm_i , Σl_i , Σm_i^2 , and $\Sigma m_i l_i$; therefore, the last two columns of Table 8.1 show the quantities m_i^2 and $m_i l_i$, and the corresponding sum is shown at the bottom of each column.

Computing the constants A and B is now straightforward. According to (8.12), the denominator Δ is

$$\begin{aligned} \Delta &= N \Sigma m^2 - (\Sigma m)^2 \\ &= 5 \times 220 - 30^2 = 200. \end{aligned}$$

Next, from (8.10) we find the intercept (the unstretched length)

$$\begin{aligned} A &= \frac{\Sigma m^2 \Sigma l - \Sigma m \Sigma ml}{\Delta} \\ &= \frac{220 \times 256.6 - 30 \times 1622}{200} = 39.0 \text{ cm.} \end{aligned}$$

Finally, from (8.11) we find the slope

$$\begin{aligned} B &= \frac{N \Sigma ml - \Sigma m \Sigma l}{\Delta} \\ &= \frac{5 \times 1622 - 30 \times 256.6}{200} = 2.06 \text{ cm/kg.} \end{aligned}$$

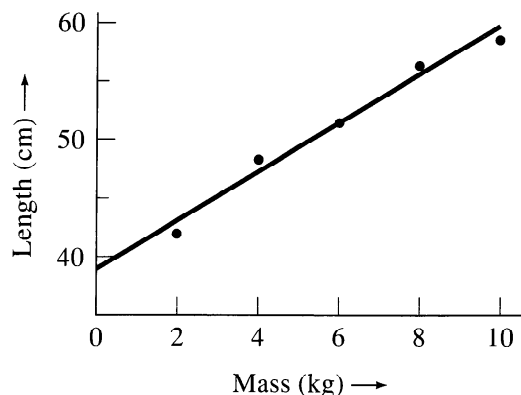


Figure 8.2. A plot of the data from Table 8.1 and the best-fit line (8.13).

A plot of the data and the line (8.13) using these values of A and B is shown in Figure 8.2. If the mass m stretches the spring to 53.2 cm, then according to (8.13) the mass is

$$m = \frac{l - A}{B} = \frac{(53.2 - 39.0) \text{ cm}}{2.06 \text{ cm/kg}} = 6.9 \text{ kg}.$$

Quick Check 8.1. Find the least-squares best-fit line $y = A + Bx$ through the three points $(x, y) = (-1, 0)$, $(0, 6)$, and $(1, 6)$. Using squared paper, plot the points and your line. [Note that because the three values of x (-1 , 0 , and 1) are symmetric about zero, $\sum x = 0$, which simplifies the calculation of A and B . In some experiments, the values of x can be arranged to be symmetrically spaced in this way, which saves some trouble.]

Now that we know how to find the best estimates for the constants A and B , we naturally ask for the uncertainties in these estimates. Before we can find these uncertainties, however, we must discuss the uncertainty σ_y in the original measurements of $y_1, y_2, \dots, y_N$.

8.3 Uncertainty in the Measurements of y

In the course of measuring the values $y_1, \dots, y_N$, we have presumably formed some idea of their uncertainty. Nonetheless, knowing how to calculate the uncertainty by analyzing the data themselves is important. Remember that the numbers $y_1, \dots, y_N$ are *not* N measurements of the same quantity. (They might, for instance, be the times for a stone to fall from N different heights.) Thus, we certainly do not get an idea of their reliability by examining the spread in their values.

Nevertheless, we can easily estimate the uncertainty σ_y in the numbers $y_1, \dots, y_N$. The measurement of each y_i is (we are assuming) normally distributed about its true value $A + Bx_i$, with width parameter σ_y . Thus the *deviations* $y_i - A - Bx_i$ are

normally distributed, all with the same central value zero and the same width σ_y . This situation immediately suggests that a good estimate for σ_y would be given by a sum of squares with the familiar form

$$\sigma_y = \sqrt{\frac{1}{N} \sum (y_i - A - Bx_i)^2}. \quad (8.14)$$

In fact, this answer can be confirmed by means of the principle of maximum likelihood. As usual, the best estimate for the parameter in question (σ_y here) is that value for which the probability (8.4) of obtaining the observed values $y_1, \dots, y_N$ is maximum. As you can easily check by differentiating (8.4) with respect to σ_y and setting the derivative equal to zero, this best estimate is precisely the answer (8.14). (See Problem 8.12.)

Unfortunately, as you may have suspected, the estimate (8.14) for σ_y is not quite the end of the story. The numbers A and B in (8.14) are the unknown true values of the constants A and B . In practice, these numbers must be replaced by our *best estimates* for A and B , namely, (8.10) and (8.11), and this replacement slightly reduces the value of (8.14). It can be shown that this reduction is compensated for if we replace the factor N in the denominator by $(N - 2)$. Thus, our final answer for the uncertainty in the measurements $y_1, \dots, y_N$ is

$$\sigma_y = \sqrt{\frac{1}{N - 2} \sum_{i=1}^N (y_i - A - Bx_i)^2}, \quad (8.15)$$

with A and B given by (8.10) and (8.11). If we already have an independent estimate of our uncertainty in $y_1, \dots, y_N$, we would expect this estimate to compare with σ_y as computed from (8.15).

I will not attempt to justify the factor of $(N - 2)$ in (8.15) but can make some comments. First, as long as N is moderately large, the difference between N and $(N - 2)$ is unimportant anyway. Second, that the factor $(N - 2)$ is *reasonable* becomes clear if we consider measuring just two pairs of data (x_1, y_1) and (x_2, y_2) . With only two points, we can always find a line that passes *exactly* through both points, and the least-squares fit will give this line. That is, with just two pairs of data, we cannot possibly deduce anything about the reliability of our measurements. Now, since both points lie exactly on the best line, the two terms of the sum in (8.14) and (8.15) are zero. Thus, the formula (8.14) (with $N = 2$ in the denominator) would give the absurd answer $\sigma_y = 0$; whereas (8.15), with $N - 2 = 0$ in the denominator, gives $\sigma_y = 0/0$, indicating correctly that σ_y is undetermined after only two measurements.

The presence of the factor $(N - 2)$ in (8.15) is reminiscent of the $(N - 1)$ that appeared in our estimate of the standard deviation of N measurements of one quantity x , in Equation (5.45). There, we made N measurements $x_1, \dots, x_N$ of the one quantity x . Before we could calculate σ_x , we had to use our data to find the mean $\bar{x}$. In a certain sense, this computation left only $(N - 1)$ independent measured values, so we say that, having computed $\bar{x}$, we have only $(N - 1)$ *degrees of freedom* left. Here, we made N measurements, but before calculating σ_y we had to compute the *two* quantities A and B . Having done this, we had only $(N - 2)$ degrees of

freedom left. In general, we define the *number of degrees of freedom* at any stage in a statistical calculation as the number of independent measurements *minus* the number of parameters calculated from these measurements. We can show (but will not do so here) that the number of degrees of freedom, *not* the number of measurements, is what should appear in the denominator of formulas such as (8.15) and (5.45). This fact explains why (8.15) contains the factor $(N - 2)$ and (5.45) the factor $(N - 1)$.

8.4 Uncertainty in the Constants A and B

Having found the uncertainty σ_y in the measured numbers $y_1, \dots, y_N$, we can easily return to our estimates for the constants A and B and calculate their uncertainties. The point is that the estimates (8.10) and (8.11) for A and B are well-defined functions of the measured numbers $y_1, \dots, y_N$. Therefore, the uncertainties in A and B are given by simple error propagation in terms of those in $y_1, \dots, y_N$. I leave it as an exercise for you to check (Problem 8.16) that

$$\sigma_A = \sigma_y \sqrt{\frac{\sum x^2}{\Delta}} \quad (8.16)$$

and

$$\sigma_B = \sigma_y \sqrt{\frac{N}{\Delta}} \quad (8.17)$$

where Δ is given by (8.12) as usual.

The results of this and the previous two sections were based on the assumptions that the measurements of y were all equally uncertain and that any uncertainties in x were negligible. Although these assumptions often are justified, we need to discuss briefly what happens when they are not. First, if the uncertainties in y are not all equal, we can use the method of *weighted least squares*, as described in Problem 8.9. Second, if there are uncertainties in x but not in y , we can simply interchange the roles of x and y in our analysis. The remaining case is that in which both x and y have uncertainties—a case that certainly *can* occur. The least-squares fitting of a general curve when both x and y have uncertainties is rather complicated and even controversial. In the important special case of a *straight line* (which is all we have discussed so far), uncertainties in both x and y make surprisingly little difference, as we now discuss.

Suppose, first, that our measurements of x are subject to uncertainty but those of y are not, and we consider a particular measured point (x, y) . This point and the true line $y = A + Bx$ are shown in Figure 8.3. The point (x, y) does not lie on the line because of the error—call it Δx —in our measurement of x . Now, we can see

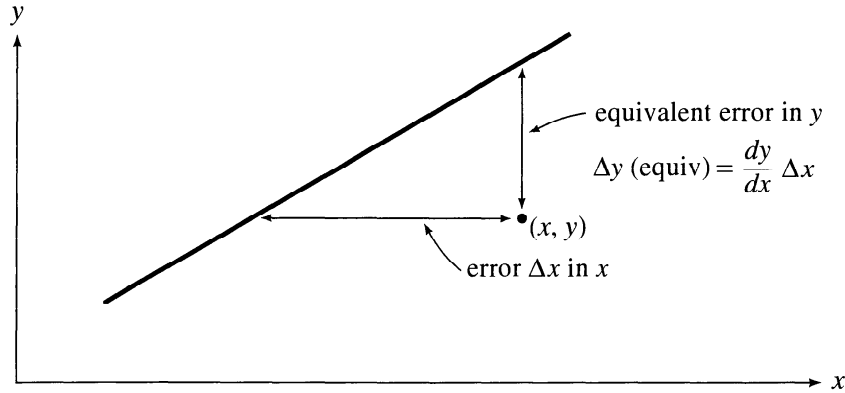


Figure 8.3. A measured point (x, y) and the line $y = A + Bx$ on which the point is supposed to lie. The error Δx in x , with y exact, produces the same effect as an error $\Delta y(\text{equiv}) = (dy/dx)\Delta x$ in y , with x exact. (Here, dy/dx denotes the slope of the expected line.)

easily from the picture that the error Δx in x , with no error in y , produces exactly the same effect as if there had been no error in x but an error in y given by

$$\Delta y(\text{equiv}) = \frac{dy}{dx} \Delta x \quad (8.18)$$

(where “equiv” stands for “equivalent”). The standard deviation σ_x is just the root-mean-square value of Δx that would result from repeating this measurement many times. Thus, according to (8.18), the problem with uncertainties σ_x in x can be replaced with an equivalent problem with uncertainties in y , given by

$$\sigma_y(\text{equiv}) = \frac{dy}{dx} \sigma_x. \quad (8.19)$$

The result (8.19) is true whatever the curve of y vs x , but (8.19) is especially simple if the curve is a straight line, because the slope dy/dx is just the constant B . Therefore, for a straight line

$$\sigma_y(\text{equiv}) = B\sigma_x. \quad (8.20)$$

In particular, if all the uncertainties σ_x are equal, the same is true of the equivalent uncertainties $\sigma_y(\text{equiv})$. Therefore, the problem of fitting a line to points (x_i, y_i) with equal uncertainties in x but no uncertainties in y is equivalent to the problem of equal uncertainties in y but none in x . This equivalence means we can safely use the method already described for either problem. [In practice, the points do not lie *exactly* on the line, and the two “equivalent” problems will not give *exactly* the same line. Nevertheless, the two lines should usually agree within the uncertainties given by (8.16) and (8.17). See Problem 8.17.]

We can now extend this argument to the case that *both* x and y have uncertainties. The uncertainty in x is equivalent to an uncertainty in y as given by (8.20). In addition, y is already subject to its own uncertainty σ_y . These two uncertainties are

independent and must be combined in quadrature. Thus, the original problem, with uncertainties in both x and y , can be replaced with an equivalent problem in which only y has uncertainty, given by

$$\sigma_y(\text{equiv}) = \sqrt{\sigma_y^2 + (B\sigma_x)^2}. \quad (8.21)$$

Provided all the uncertainties σ_x are the same, and likewise all the uncertainties σ_y , the equivalent uncertainties (8.21) are all the same, and we can safely use the formulas (8.10) through (8.17).

If the uncertainties in x (or in y) are not all the same, we can still use (8.21), but the resulting uncertainties will not all be the same, and we will need to use the method of weighted least squares. If the curve to which we are fitting our points is not a straight line, a further complication arises because the slope dy/dx is not a constant and we cannot replace (8.19) with (8.20). Nevertheless, we can still use (8.21) (with dy/dx in place of B) to replace the original problem (with uncertainties in both x and y) by an equivalent problem in which only y has uncertainties as given by (8.21).¹

8.5 An Example

Here is a simple example of least-squares fitting to a straight line; it involves the constant-volume gas thermometer.

Example: Measurement of Absolute Zero with a Constant-Volume Gas Thermometer

If the volume of a sample of an ideal gas is kept constant, its temperature T is a linear function of its pressure P ,

$$T = A + BP. \quad (8.22)$$

Here, the constant A is the temperature at which the pressure P would drop to zero (if the gas did not condense into a liquid first); it is called the *absolute zero of temperature*, and has the accepted value

$$A = -273.15^\circ\text{C}$$

The constant B depends on the nature of the gas, its mass, and its volume.² By measuring a series of values for T and P , we can find the best estimates for the constants A and B . In particular, the value of A gives the absolute zero of temperature.

One set of five measurements of P and T obtained by a student was as shown in the first three columns of Table 8.2. The student judged that his measurements of

¹This procedure is quite complicated in practice. Before we can use (8.21) to find the uncertainty $\sigma_y(\text{equiv})$, we need to know the slope B (or, more generally, dy/dx), which is not known until we have solved the problem! Nevertheless, we can get a reasonable first approximation for the slope using the method of unweighted least squares, ignoring all of the complications discussed here. This method gives an approximate value for the slope B , which can then be used in (8.21) to give a reasonable approximation for $\sigma_y(\text{equiv})$.

²The difference $T - A$ is called the *absolute temperature*. Thus (8.22) can be rewritten to say that the absolute temperature is proportional to the pressure (at constant volume).